

AMTI (NMTC) - 2004
BHASKARA CONTEST - FINAL - JUNIOR LEVEL

1. Show that there are no integers a, b, c for which $a^2 + b^2 - 8c = 6$.
2. Given that $N = 2^n (2^{n+1} - 1)$ and $2^{n+1} - 1$ is a prime number, show that
 - a) Sum of the divisors of N is $2N$
 - b) Sum of the reciprocals of the divisors of N is 2 .
3. Given three non-collinear points A, B, C construct a circle with center C such that the tangents from A and B to the circle are parallel.
4. Given a circle with diameter AB and a point X on the circle different from A and B , let t_a , t_b and t_x be the tangents to the circle at A, B and X respectively. Let Z be the point where the line AX meets t_b and Y be the point where the line BX meets t_a . Show that the three lines YZ , t_x and AB are either concurrent or parallel.
5. The polynomial $ax^3 + bx^2 + cx + d$ has integral coefficients a, b, c, d . If ad is odd and bc is even show that at least one root of the polynomial is irrational.
6. Let f be a function from N to R satisfying
 - (a) $f(1) = 1$ and (b) $f(1) + 2f(2) + 3f(3) + \dots + nf(n) = n(n+1)f(n)$. Find $f(2004)$.
7. Consider a permutation $p_1 p_2 p_3 p_4 p_5 p_6$ of the six numbers $1, 2, 3, 4, 5, 6$ which can be transformed to 123456 by transposing two numbers exactly four times. By a transposition we mean an interchange of two places – for example, $1\ 2\ 3\ 4\ 5\ 6$ to $3\ 2\ 1\ 4\ 5\ 6$ (positions 1 and 3 are interchanged). Find the number of such permutations.
8. Let $a_1, a_2, a_3, \dots, a_m$ be a sequence of real numbers. The sum of k - successive terms is called a k - sum, for example $a_j + a_{j+1} + a_{j+2} + \dots + a_{j+k-1}$ is a k - sum. In a finite sequence of real numbers every 7 -sum is negative and every 11 -sum is positive. Find the largest number of terms in such a sequence.
